Conjecture 00000001559: the six-disk covering radius of the unit disk

Disproof of the stated value of r_6

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1 The conjecture as filed

The statement, quoted in English from the filing (symbols typeset in math mode, wording unchanged):

> Definition: The n-disk covering radius r_n of the unit disk is the minimal radius of n equal disks covering it. Conjecture: r_6 is an explicit quadratic algebraic number of the $(1+\sqrt{3})/2\cdot(\sqrt{3}-1)$ type, realized by one central disk and five symmetric annular disks; and r_7 is a cubic algebraic number.

2 Precise reading

Let $D=\{p\in\mathbb{R}^2 : |p|\le 1\}$ be the closed unit disk and $B(c,r)=\{p : |p-c|\le r\}$ the closed disk of radius r centred at c. The covering radius used is

$$r_n \;=\; \inf\Big\{\, r\ge 0 \;:\; \exists\, c_1,\dots,c_n\in\mathbb{R}^2,\; D\subseteq\textstyle\bigcup_{i=1}^{n} B(c_i,r)\,\Big\}. \tag{1}$$

Centres are unrestricted. Repetitions are allowed in the main reading. The infimum is attained by compactness, so it is a minimum. The disproof also holds when the n centres must be pairwise distinct (see Section 4).

The clause "r_6 is of the $(1+\sqrt{3})/2\cdot(\sqrt{3}-1)$ type" is read as the equality

$$r_6 \;=\; \frac{1+\sqrt{3}}{2}\,(\sqrt{3}-1).$$

The right-hand side is exactly 1, because $(1+\sqrt{3})(\sqrt{3}-1)=3-1=2$. The literal claim is therefore $r_6=1$. The expression is rational, so it is not a quadratic irrational as the text describes.

3 Counterexample and verification

3.1 The certificate

Let $c_1=(\frac12,\frac12)$, $c_2=(-\frac12,\frac12)$, $c_3=(-\frac12,-\frac12)$, $c_4=(\frac12,-\frac12)$, and take six disks: these four, with c_1 and c_3 repeated. Let $p=(x,y)\in D$. Choose the corner $c=(\operatorname{sgn}(x)/2,\operatorname{sgn}(y)/2)$. Expanding gives

$$|p-c|^2 \;=\; x^2+y^2-|x|-|y|+\tfrac12. \tag{2}$$

Since $x^2+y^2 \le 1$ we have $|x| \le 1$, so $x^2 \le |x|$. Likewise $y^2 \le |y|$. Hence $x^2+y^2-|x|-|y| \le 0$, and therefore $|p-c|^2 \le \frac{1}{2}$. Every point of D is within distance $1/\sqrt{2}$ of one of the six centres, which gives

$$r_6 \;\le\; \frac{1}{\sqrt{2}} \;<\; 1.$$

This contradicts the claimed value $r_6 = 1$.

3.2 Computed values

The script `check.py` (sympy 1.14.0, numpy 2.5.4) gives the following. The full output is in `check_output.txt`.

Part	Quantity	Value
A	claimed expression $(1+\sqrt{3})/2 \cdot (\sqrt{3}-1)$, exact (sympy)	1
B	corner certificate, polar-grid maximum distance	$0.70710678 \le 1/\sqrt{2}$
B	exact rational identity (2), 15 696 points in D	holds; $d^2 \le \frac{1}{2}$
B	largest exact d^2 among the samples	$3094976921/6250000000 \approx 0.4952 \le 0.5$
C	hexagon of radius 0.7 (scanner's suggestion), grid maximum	0.7; convex bound 0.49
D	random-restart search, coarse / fine grid (upper bound only)	0.5556 / 0.5585

Part D is only a numerical upper bound, and it is not a proof of the value of r_6. It shows that r_6 is far below 1, and it is consistent with the disproof. The r_7 clause was not checked. The refutation of the r_6 clause alone is enough to falsify the statement as written.

4 Lean formalization

The file is `Disproof.lean`, in namespace `TLMC1559`. It imports Mathlib. Mathlib has no covering radius, so the objects are defined directly. Distances are written in coordinates, because the product metric on $\mathbb{R} \times \mathbb{R}$ is the sup-norm and must not be used.

```
def inDisk (p : R x R) : Prop := p.1 ^ 2 + p.2 ^ 2 <= 1

def Covers (r : R) (C : Fin 6 -> R x R) : Prop :=
  forall  p : R x R, inDisk p -> exists  i, (p.1 - (C i).1) ^ 2 + (p.2 - (C i).2) ^ 2 <=
    r ^ 2

def coverRadius : Set R :=
  {r | 0 <= r /\ exists  C : Fin 6 -> R x R, Covers r C}

def coverRadiusInj : Set R :=
  {r | 0 <= r /\ exists  C : Fin 6 -> R x R, Function.Injective C /\ Covers r C}

noncomputable def r6 : R := sInf coverRadius
noncomputable def r6Inj : R := sInf coverRadiusInj
```

The certificate is `C6` (the corners with two repeats) and `C6d` (the corners plus $(0,0)$ and $(3,3)$, which are pairwise distinct). The corner lemma `cover_of_corners` proves, by a case split on the signs of x and y, that every $p \in D$ has a centre at squared distance at most $\frac{1}{2}$. The bounds $|x|, |y| \le 1$ are obtained with `nlinarith`. The lemmas `covers_C6` and `covers_C6d` show that both certificates cover D with radius $3/4$. Then `r6_le` and `r6Inj_le` give $r_6 \le 3/4$ through

`csInf_le`, and `conjectured_expr_eq_one` shows that the expression equals 1 using $\sqrt{3}^2 = 3$ and `linear_combination`.

The main theorem is:

```
theorem conjecture_00000001559_fails :
    not (r6 = (1 + Real.sqrt 3) / 2 * (Real.sqrt 3 - 1)) /\
    not (r6Inj = (1 + Real.sqrt 3) / 2 * (Real.sqrt 3 - 1))
```

In the listings, ASCII stands for the Unicode symbols of the Lean source: `R` for $\mathbb{R}$, `/\` for the conjunction, `not` for negation, `->` for the arrow.

The first conjunct is the disproof with unrestricted centres. The second is the disproof with pairwise distinct centres.

Build result. Running `lake env lean` on the file gives no errors and no warnings. The line `#print axioms` reports

```
'TLMC1559.conjecture_00000001559_fails' depends on axioms: [propext, Classical.choice,
    Quot.sound]
```

The file contains no `sorry`, `native_decide`, `axiom`, `unsafe`, `opaque`, `implemented_by`, `extern`, or `admit`. The distinctness of the six points of `C6d` is checked by `fin_cases` and `norm_num`; no `decide` is used.

Scope. The formal statement refutes the stated value of r_6. It does not formalize the structural claims (a central disk with five annular disks) or the cubic claim for r_7.

5 Reproduction

1. Verifier (about two minutes, mainly the random-restart search):

   ```
   cd LibertychaserUS_submission_20261027064631
   python3 check.py
   ```

2. Lean (Lean v4.33.1, Mathlib v4.33.1):

   ```
   cd lean
   lake exe cache get \&\& lake build
   ```

3. Report:

   ```
   cd LibertychaserUS_submission_20261027064631
   tectonic report.tex
   ```